Technical Document #398: Natural Language Processing - Comprehensive Overview

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Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Coreference resolution identifies all expressions that refer to the same entity in a text. It is essential for understanding document-level coherence.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Tokenization is the process of breaking text into individual units called tokens.

- Word embeddings represent words as dense vectors in continuous space.

- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person nam